CT DOSE OPTIMISATION · PROJECT REPORT

CT Dose AI

Predicting $\mathrm{CTDI_{vol}}$ and DLP for coronary CT angiography, and learning the lowest radiation dose that still gives a diagnostic image.

0.9%
$\mathrm{CTDI_{vol}}$ prediction error (cross-validated)

8.2%
DLP prediction error (cross-validated)

≈40%
dose saving at 100 kV at equal estimated iodine CNR

50
exams used to build and validate the model

What this document covers. Sections 1–2 state the clinical problem and audit the data. Section 3 describes the algorithm: the physics-informed prediction core, the recommendation engine, and the two learning loops (Bayesian optimisation on a phantom, and a safe contextual bandit in clinical routine). Section 4 shows the delivered application, Section 5 the measured results, Section 6 the open questions for the data provider, and Section 7 limitations and next steps. Appendix A holds the mathematics, Appendix B the code map.

> **STATUS**
> Research prototype and decision-support tool. It is **not** a medical device and has not been clinically validated. Every protocol change must be reviewed by a medical physicist and the responsible radiologist before use on patients.

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Author: Thodoris Fragos · **Data:** 50 coronary CT angiography exams (CARDIAC CT, provided by a clinical colleague) · **Date:** 23 September 2026

Repository: `https://github.com/Thodoris1657/CT-Dose-AI-Algorithm` · **Stack:** Python, scikit-learn, SciPy, Streamlit, Plotly

1 · The problem

1.1 The clinical request

The project started from a request by a clinical colleague, who supplied the first batch of data:

> *"We want the AI programme (an app or a website, whichever you think) to take BMI, kVp, mAs and scan length, and predict CTDI and DLP. First for this specific examination. Then we want to try other examinations too, and to see whether — and how — it can find CTDI and DLP and suggest what I should change in kVp, mAs or scan length in order to reduce CTDI and DLP. We consider BMI fixed, because it depends on the patient's height and weight."*

That request contains two very different problems, and separating them is the key design decision of the project:

1. **A prediction problem.** Given the patient and the planned protocol, what dose will the scanner deliver? This is a supervised-learning problem with a strong physical structure.
2. **A decision problem.** Which settings should change to lower the dose? A dose-only answer is trivial and dangerous: the dose is minimised by scanning with almost no radiation, which produces useless images. The real question is constrained — *the lowest dose that still yields a diagnostic image*. That constraint depends on image quality, which is not present in the supplied data at all.

1.2 Why it matters

Coronary CT angiography is one of the highest-dose routine CT examinations, and it is performed on patients who are often young enough for stochastic risk to matter. National and European reference levels for CCTA are of the order of 170–400 mGy·cm for prospectively triggered protocols and 870–1060 mGy·cm for retrospectively gated ones [1,2]. The 50 supplied exams have a median DLP of about 1080 mGy·cm, with a maximum of 2393 mGy·cm.

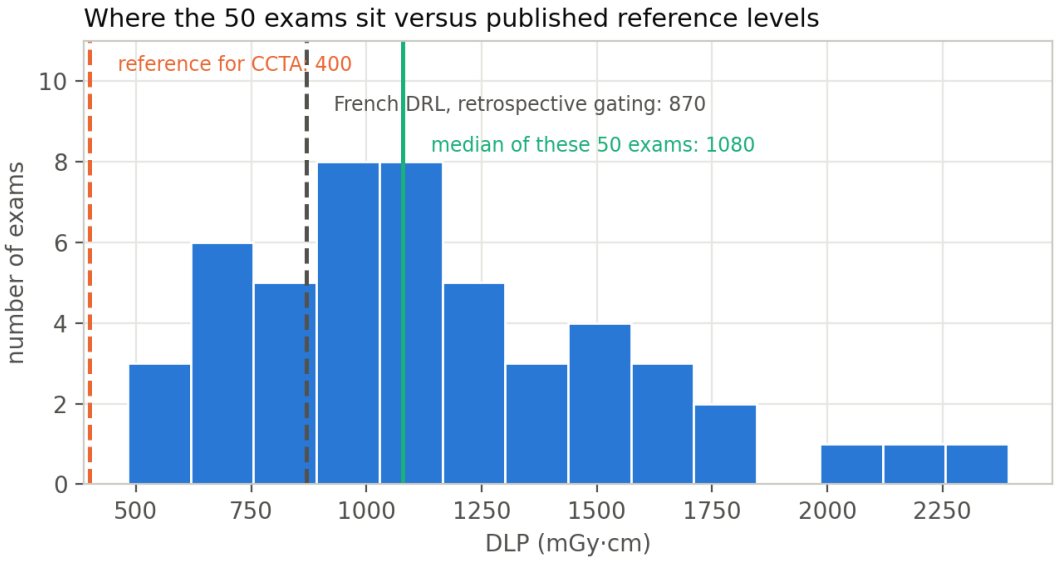


Figure 1. DLP of the 50 supplied exams. The dose level is consistent with retrospectively gated acquisition at 120 kV. Almost every exam sits above the 400 mGy·cm reference proposed for CCTA, which is where the opportunity lies: there is real headroom to reduce dose, provided image quality is protected.

1.3 What makes this hard

- **Small data.** 50 exams, of which 48 are unique. Any model with many free parameters will memorise noise. Validation has to be honest, or the project will produce an impressive-looking number and a useless tool.
- **Narrow input range.** The data contains only 120 kV (46 exams) and 140 kV (4 exams). The most effective dose-reduction lever — dropping to 100 kV or 80 kV — lies *outside* the observed data. A purely statistical model cannot extrapolate there; a physics-based one can, with a stated assumption.
- **No image-quality signal.** Nothing in the data says whether an image was diagnostic. A recommendation engine built on dose alone would simply recommend "less".

- **BMI is fixed.** The one variable most correlated with dose is the one that cannot be changed. All the levers are protocol parameters.

2 · The data and what it revealed

The source file is a spreadsheet printed to PDF (`CARDIAC CT KARANISA.pdf`), with the columns split across three blocks and six pages. The application parses it directly, so the same PDF can be re-imported at any time; CSV and Excel are also supported.

Field	Role	Median	Range	Comment
Examination	grouping		CCTA, all 50	one model per exam type
BMI (kg/m^2)	input, fixed	27.2	18.3–58.8	patient property
kVp	input, tunable	120	120 (n=46), 140 (n=4)	only two levels present
mAs	input, tunable	756	375–1345	set by tube-current modulation
Scan size (cm)	input, tunable	29.8	22.4–44.2	see the anomaly below
$\mathrm{CTDI_{vol}}$ (mGy)	target	47.5	21.1–105	scanner console value
DLP (mGy·cm)	target	1080	482–2393	scanner console value

2.1 The automated data audit

Before any model is fitted, the pipeline audits the data (`ctdose/data.py`). Four findings came out, and all four are clinically informative.

Finding 1 — $\mathrm{CTDI_{vol}}$ is a deterministic function of mAs and kV

$\mathrm{CTDI_{vol}}$/mAs is essentially constant: 0.0605 mGy/mAs at 120 kV and 0.0894 mGy/mAs at 140 kV. This is not a statistical correlation, it is the scanner's dose output per unit of tube charge. The ratio between the two kV levels, 1.478, implies an exponent of 2.53 in $(\mathrm{kV}/120)^n$, squarely inside the physically expected 2.5–3.0 range. This single observation determines the entire modelling strategy.

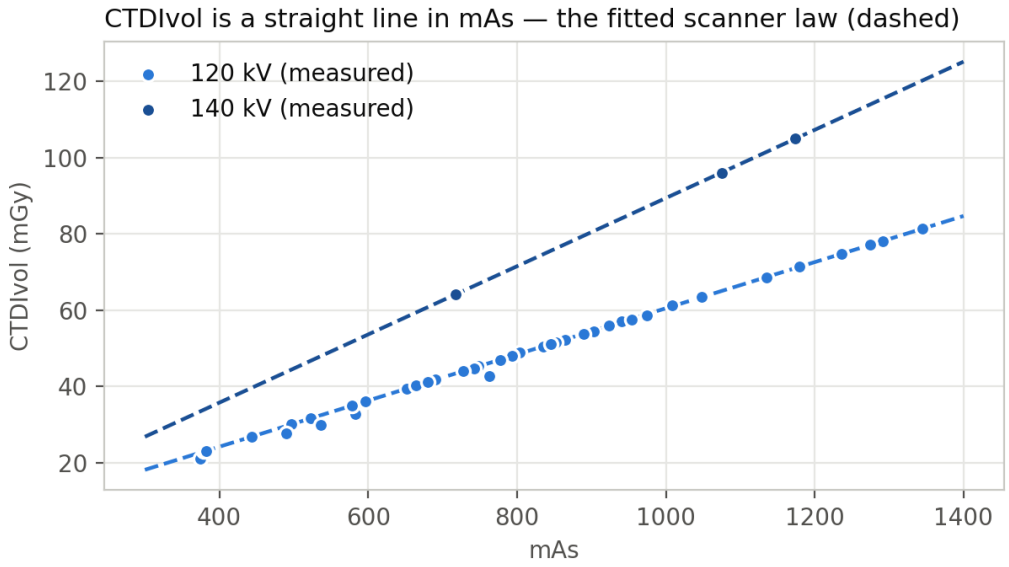


Figure 2. Measured $\mathrm{CTDI_{vol}}$ against mAs, separated by tube voltage. The dashed lines are the fitted law, not a regression through a cloud of points. The five points visibly below the 120 kV line are Finding 3.

Finding 2 — the "SCAN SIZE CM" field does not behave like the irradiated length

DLP is, by definition, $\mathrm{CTDI_{vol}}$ multiplied by the irradiated length. Dividing the two gives an implied length of 23.2 ± 2.2 cm. That implied length has *no* positive correlation with the supplied scan-size field ($r = -0.23$), while the field itself correlates strongly with BMI ($r = +0.89$). The field therefore behaves like a patient dimension or a field of view, not like z-coverage. This matters: it is the single largest source of error in the DLP model.

Finding 3 — five exams have a different dose output per mAs

Rows 2, 9, 22, 41 and 45 deliver about 7% less $\mathrm{CTDI}_{\mathrm{vol}}$ per mAs than the other 120 kV exams, and they cluster tightly, which points to a systematic cause — a different collimation, pitch or scan mode — rather than noise. Nothing in the four supplied inputs identifies them, so they set a floor on achievable accuracy.

Finding 4 — two pairs of duplicate rows

Rows 28/43 and 29/42 are identical in every field. The pipeline keeps duplicates inside the same cross-validation fold, so they cannot leak from training into testing and flatter the reported accuracy.

3 · The algorithm

The system has one physics core and three layers built on it. The prediction layer answers *what dose will this protocol give?*; the recommendation layer answers *what should I change?*; and two learning loops close the gap the data cannot fill — *how low can we actually go?*

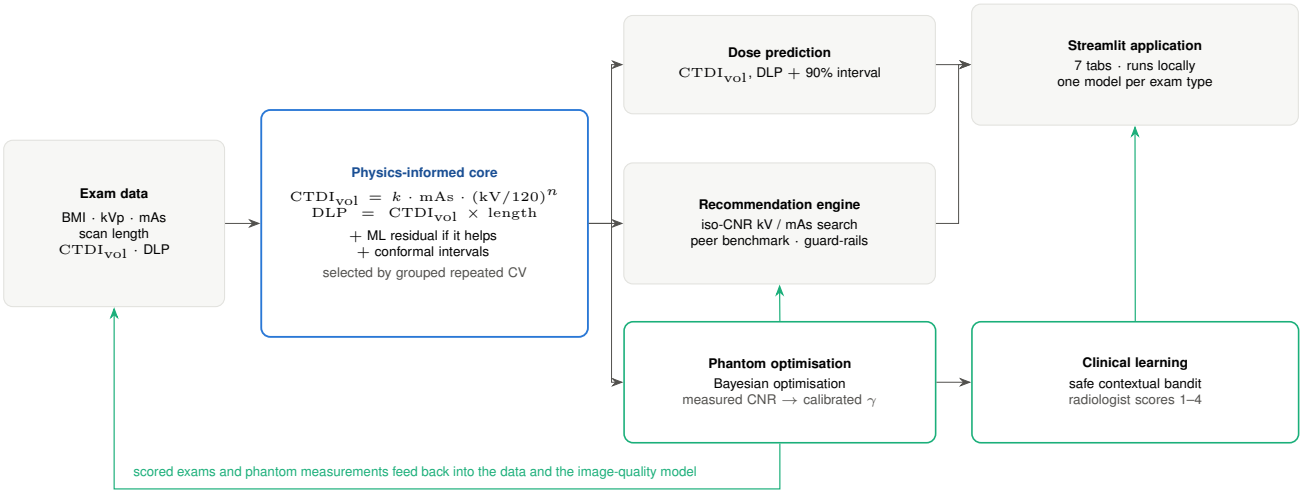


Figure 3. System architecture. Blue = the dose model built from the supplied data; green = the two loops that acquire the image-quality information the data does not contain.

3.1 Physics-informed prediction core

All models work in log space, because dose quantities are positive and their drivers multiply. The physical law for a given scanner is:

$$\mathrm{CTDI_{vol}} = k \cdot \mathrm{mAs} \cdot (\mathrm{kVp}/120)^{n} \implies \log \mathrm{CTDI_{vol}} = \log k + \mathbf{1} \cdot \log \mathrm{mAs} + n \log(\mathrm{kVp}/120)$$
$$\mathrm{DLP} = \mathrm{CTDI_{vol}} \times L_{\text{irradiated}} \implies \log \mathrm{DLP} = \log \mathrm{CTDI_{vol}} + \log L$$

Two modelling choices follow, and both are deliberate:

- **The mAs exponent is fixed at 1**, not fitted. Dose is linear in tube charge by definition. Fixing it means the model extrapolates safely to mAs values never seen in training, and removes one parameter from a 48-row dataset.
- **The fit is robust (Huber loss)**, so the five deviant exams from Finding 3 do not drag the fitted constant away from the behaviour of the other 45.

Fitted values for this scanner: $k = 0.0605$ mGy/mAs and $n = 2.53$. The DLP model adds a fitted effective length, which absorbs the fact that the true z-coverage is not recorded.

3.2 Model competition and validation

The physics law was not assumed to be best — it had to win a competition against eight alternatives, including black-box models, under an identical protocol:

- **Candidates:** physics law; physics + BMI; physics + length; physics + Gaussian-process residual; physics + neural-network residual; physics + gradient-boosting residual; log–log ridge regression with interactions; gradient boosting with monotonicity constraints on mAs and kV; and a bagged ensemble of deep multilayer perceptrons (the “deep learning” entry).
- **Protocol:** repeated grouped 5-fold cross-validation, 10 repeats (50 fits per model). Duplicate rows are forced into the same fold.
- **Selection:** the 1-standard-error rule — the simplest model whose error is within one standard error of the best. This prevents a complex model winning on a fluctuation.

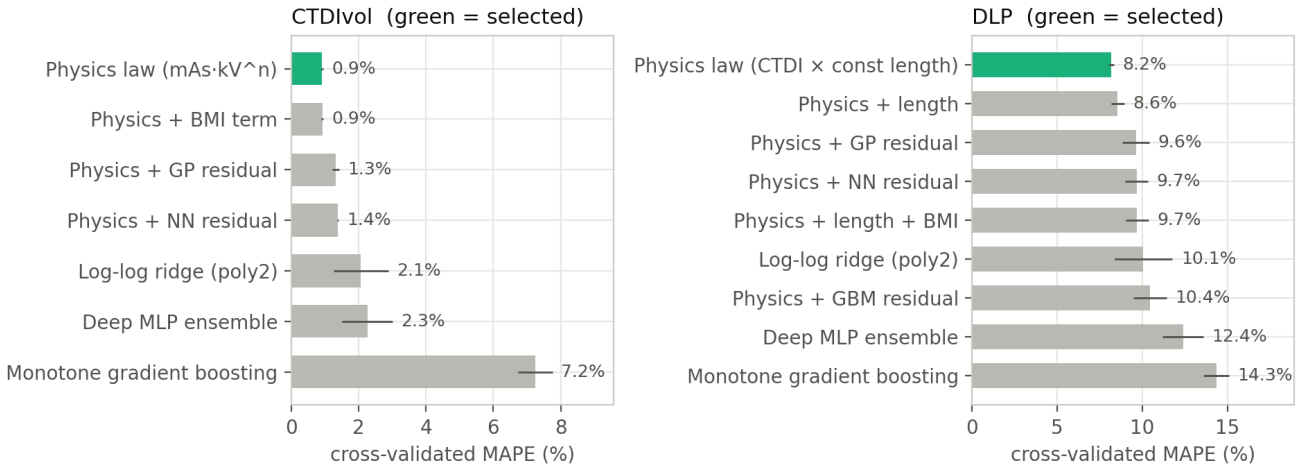


Figure 4. Cross-validated mean absolute percentage error, with the spread across repeats. For both targets the physics-informed model wins outright. The deep ensemble is 2.5× worse on $\mathrm{CTDI_{vol}}$ and 1.5× worse on DLP — the expected outcome with 48 unique rows and a known functional form.

Model	$\mathrm{CTDI_{vol}}$ error	R^2	DLP error	R^2
Physics law (selected)	**0.92% ± 0.03**	**0.998**	**8.17% ± 0.17**	**0.922**
Physics + Gaussian-process residual	1.32%	0.998	9.63%	0.854
Physics + neural-network residual	1.38%	0.998	9.66%	0.849
Log–log ridge with interactions	2.07%	0.960	10.07%	0.734
Deep MLP ensemble	2.26%	0.964	12.39%	0.586
Monotone gradient boosting	7.25%	0.838	14.35%	0.695

WHY THE SIMPLE MODEL IS THE RIGHT ANSWER, NOT A SHORTCUT

A deep network is the correct tool when the functional form is unknown and data is plentiful. Here the form is known exactly and the data is scarce, so the network spends its capacity re-learning a straight line from 48 points — and does it worse. The value added by machine learning in this project is not in the function approximator; it is in the validation protocol, the calibrated uncertainty, the constrained optimisation, and the two active-learning loops. Should the dataset grow to thousands of exams with heart rate, pitch, gating mode and reconstruction level, the same script re-runs the full competition and will select a flexible model automatically if one then wins.

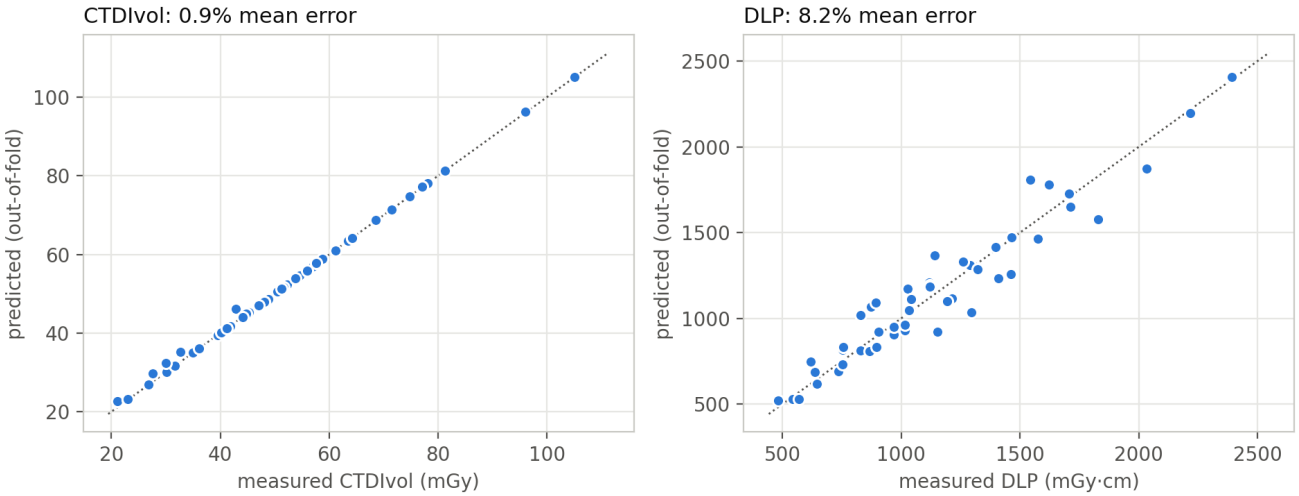


Figure 5. Out-of-fold predictions (every point predicted by a model that never saw it). $\mathrm{CTDI_{vol}}$ lies on the identity line except for the five deviant exams; the DLP scatter reflects the unrecorded scan length.

3.3 Uncertainty and applicability domain

A point prediction without an error bar invites misuse, so every prediction carries a **conformal prediction interval** calibrated on out-of-fold residuals. Conformal intervals are distribution-free: they guarantee the stated coverage under exchangeability, without assuming normal errors.

Target	90% interval (multiplicative)	Coverage	Interpretation
$\mathrm{CTDI_{vol}}$	prediction $\times\ [0.927,\ 1.003]$	94%	asymmetric: the lower tail is the 7% deviant cluster
DLP	prediction $\times\ [0.820,\ 1.248]$	94%	width reflects the unknown scan length

In addition, every prediction is checked against the **applicability domain**. Inputs outside the training ranges are flagged, and any kV value not present in the data (80, 90, 100 kV) produces an explicit warning that the estimate relies on the fitted physics kV-law rather than on observed exams. The tool states when it is extrapolating instead of hiding it.

3.4 Dose-reduction recommendation engine

Dose alone is not a valid objective, so the engine optimises dose *subject to an image-quality constraint*. For the same patient, quantum noise falls with the square root of dose, while iodine contrast rises as the tube voltage approaches the iodine K-edge:

$$\sigma_{\text{noise}} \propto 1/\sqrt{\mathrm{CTDI_{vol}}} \qquad C_{\text{iodine}} \propto \mathrm{kVp}^{-\gamma}, \quad \gamma \approx 1.2\text{–}1.5$$

$$\mathrm{CNR_{rel}} = \left(\frac{\mathrm{kVp_{old}}}{\mathrm{kVp_{new}}}\right)^{\gamma} \sqrt{\frac{D_{\mathrm{new}}}{D_{\mathrm{old}}}} \qquad \text{hold } \mathrm{CNR} \geq \tau: \quad D_{\mathrm{new}} \geq D_{\mathrm{old}}\, \tau^2 \left(\frac{\mathrm{kVp_{new}}}{\mathrm{kVp_{old}}}\right)^{2\gamma}$$

(D denotes $\mathrm{CTDI_{vol}}$; τ is the fraction of the current CNR to preserve.)

This is why lowering kV is the dominant lever in contrast-enhanced CT: at 100 kV the same iodine CNR needs about 40% less dose than at 120 kV. The engine evaluates every available kV, computes the mAs that preserves the chosen fraction of the current CNR, and ranks the results. Four guard-rails constrain the search:

- **Minimum kV by BMI** (default 80 kV below BMI 25, 100 kV for 25–30, 120 kV above 30), following PROTECTION VI [1] — this prevents photon starvation in large patients.
- **Tube output limits**: options that would exceed the tube's maximum mAs are marked as unable to reach the CNR target.
- **Peer benchmark**: quantile regression of the local data gives the median and 25th-percentile dose actually used for patients of the same BMI — an "achievable dose" that needs no extrapolation at all.
- **Scan length**: DLP is proportional to irradiated length, so the saving per centimetre is reported separately, with the caveat from Finding 2 attached.

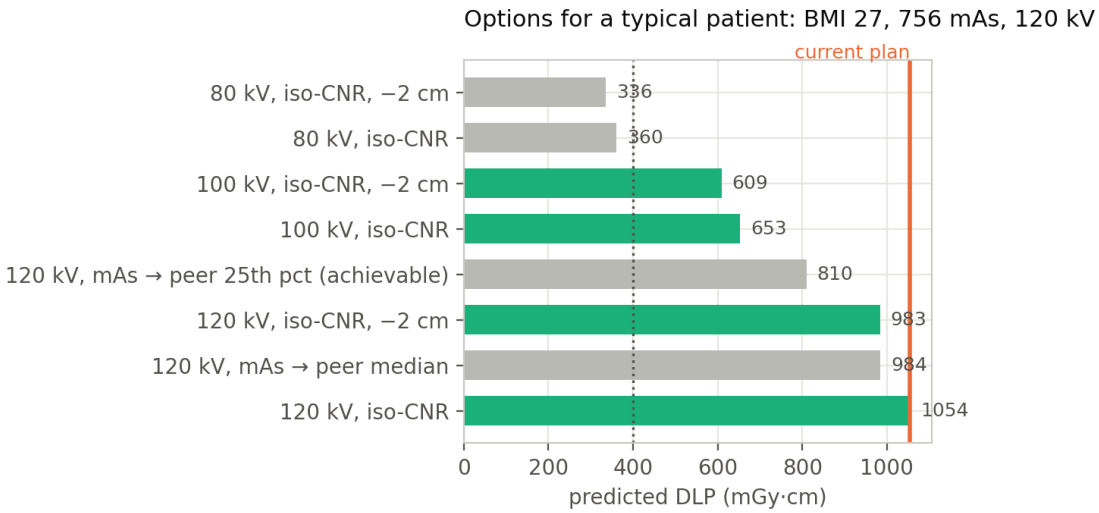


Figure 6. Ranked options for a median patient (BMI 27, 756 mAs, 120 kV; predicted 45.7 mGy and 1056 mGy·cm). Green options satisfy every constraint. The recommendation is 100 kV / 735 mAs, which is -39% $\mathrm{CTDI}_{\mathrm{vol}}$ and -42% DLP with 2 cm less coverage, at the same estimated iodine CNR. The 80 kV options are greyed out because the patient's BMI exceeds the 80 kV rule.

3.5 Phantom Bayesian optimisation

The CNR model above uses a literature exponent. The honest way to replace it with a measured one is a phantom experiment — but scanning a full grid of kV $\times$ mAs combinations wastes scanner time. Bayesian optimisation solves exactly this: choose each next measurement so that it is maximally informative about the optimum.

- **Objective:** minimise $\mathrm{CTDI_{vol}}$ subject to measured CNR $\geq$ target. Dose is known analytically from the fitted scanner law; CNR is the unknown, noisy, expensive function.
- **Surrogate:** $\log \mathrm{CNR} = b_0 - \gamma \log(\mathrm{kV}/120) + \delta \log \mathrm{CTDI_{vol}}$, fitted as a Bayesian linear model with physics priors ($\gamma \approx 1.35 \pm 0.5$, $\delta \approx 0.5 \pm 0.15$), plus a Gaussian-process residual that captures what the formula misses, such as photon starvation at low kV. The priors are what make the method useful after 2–3 scans instead of 30.
- **Acquisition:** constrained expected improvement, $a(x) = \max(0,\ d^* - d(x)) \cdot P(\mathrm{CNR}(x) \geq \text{target})$. Before any feasible point exists it takes the lowest-dose setting that is more likely than not to pass.
- **Output:** the next scan to perform, the recommended protocol, and a **calibrated** γ that replaces the literature value in the recommendation engine.

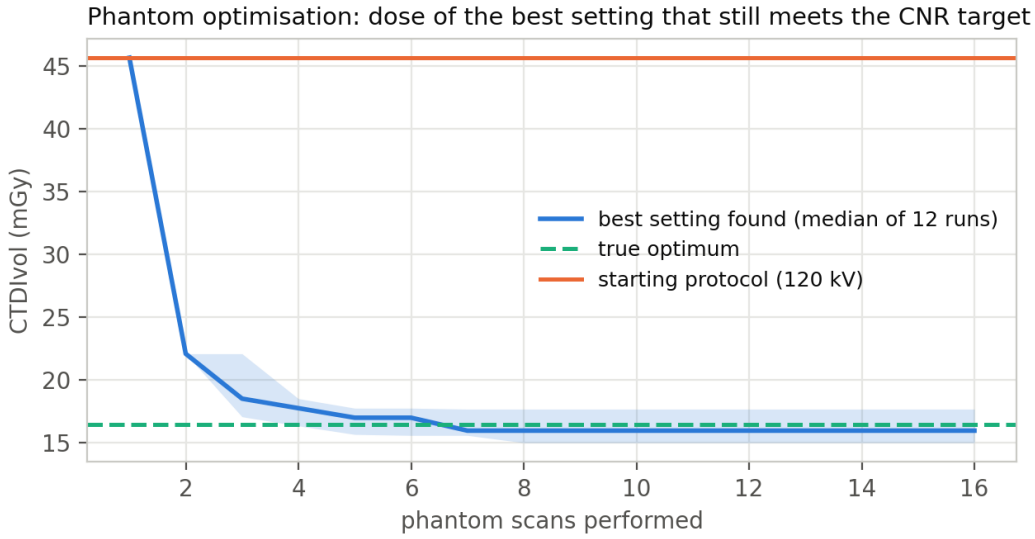


Figure 7. Simulated campaigns on a synthetic phantom (12 runs, band = 10th–90th percentile). The starting protocol is the current clinical one. After about 7 scans the procedure sits at the true constrained optimum — roughly 65% below the starting dose in this simulation. The curve can dip marginally below the true optimum because a single noisy measurement may pass the target by chance, which is why the reported recommendation uses the 95% posterior probability rather than the best single measurement.

3.6 Safe contextual bandit in clinical routine

A phantom is not a patient. The final loop learns from clinical practice itself, using the one signal that defines success: whether the radiologist found the image diagnostic.

- **Context:** the patient's BMI. **Actions:** combinations of kV and image-quality level, converted into concrete mAs through the scanner law. **Reward:** a 1–4 diagnostic-quality score, with ≥ 3 counted as acceptable.
- **Model:** Bayesian logistic regression (Laplace approximation) of $P(\text{acceptable})$ on BMI, $\log \mathrm{CTDI_{vol}}$ and $\log$ kV, with physics-informed priors — more dose helps, lower kV helps iodine contrast at equal dose, a larger patient is harder. A single shared model means every exam informs every arm, which is far more data-efficient than learning each protocol independently.
- **Decision:** Thompson sampling. One plausible model is drawn from the posterior, and the lowest-dose allowed action whose sampled $P(\text{acceptable})$ clears the target (default 0.97) is proposed. Exploration comes from the posterior draw and shrinks automatically as evidence accumulates.

> **FOUR SAFETY RAILS, ALL ACTIVE SIMULTANEOUSLY**
>
> 1. The BMI minimum-kV rule always applies.

2. The pessimistic (10th-percentile) posterior $P(\text{acceptable})$ of a proposal must exceed 0.85.
3. A proposal may be at most 15% below the lowest dose already *proven* acceptable for a similar BMI (± 3 BMI units), so the dose descends in small, evidenced steps rather than jumping.
4. A radiologist approves or overrides every single proposal, and the override is recorded.

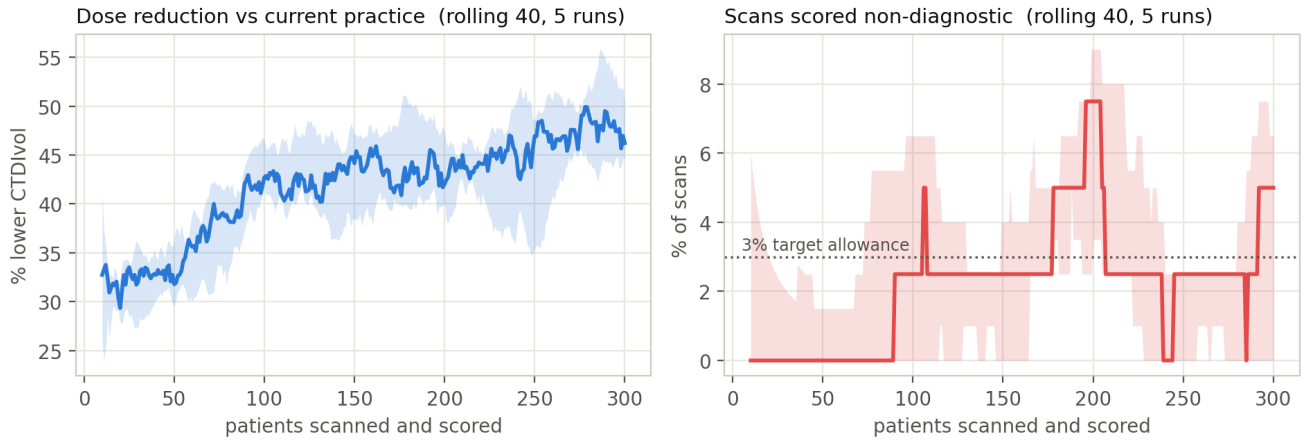


Figure 8. Simulation with a synthetic reader whose acceptance depends on true iodine CNR (5 runs $\times$ 300 patients; line $=$ median, band $=$ 10th–90th percentile). Dose falls about 32% immediately — this comes from the kV switch, which the physics prior already considers safe — and reaches 45–50% as evidence accumulates. The non-diagnostic rate stays near the 3% allowance implied by the 0.97 target, which is the parameter the department sets.

4 · The delivered solution

Everything above ships as a local Python application: `python app.py` opens it in a browser. Nothing leaves the computer, which matters for patient data. Seven tabs cover the workflow.

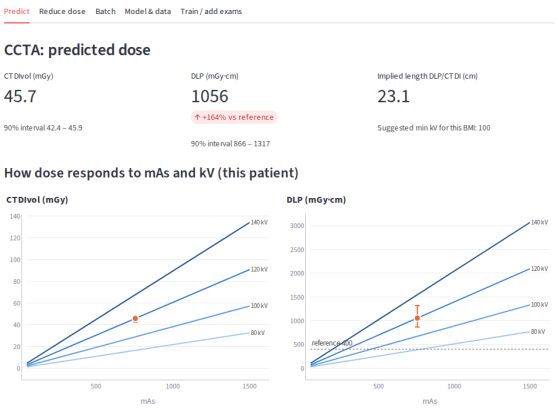


Figure 9. Predict — dose with 90% intervals, comparison against the reference level, and the dose response to mAs at each kV for this patient.

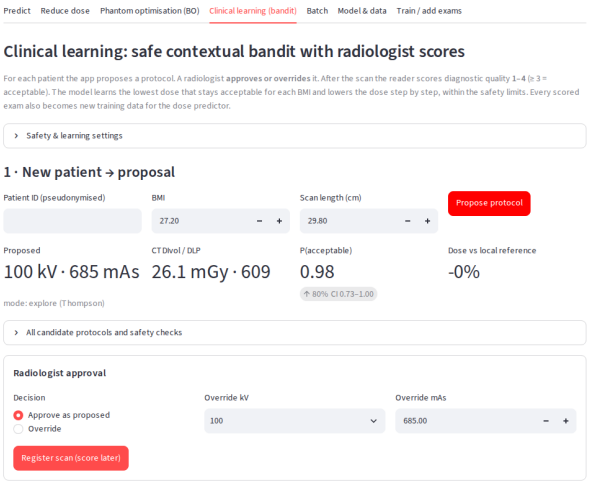


Figure 10. Clinical learning — proposal, probability of acceptability with its credible interval, approval control, and the learned curves.

Tab	What it does	Who uses it
Predict	$\mathrm{CTDI_{vol}}$ and DLP with 90% intervals, reference-level comparison, dose-response curves, out-of-domain warnings	radiographer, physicist
Reduce dose	Ranked protocol options at constant estimated CNR, peer benchmark, guard-rails, manual what-if panel	physicist, radiologist
Phantom optimisation	Proposes the next phantom scan, accepts the measured CNR, returns the optimal protocol and calibrated γ	physicist
Clinical learning	Per-patient proposal → approval → 1–4 score → model update; dashboards and export	radiologist
Batch	Predicts a whole CSV/Excel/PDF file and reports errors when true doses are present	physicist, audit
Model & data	Model card: cross-validation table, fitted law, out-of-fold plots, full data audit	reviewer, physicist
Train / add exams	Upload another exam type; it gets its own model and report automatically	anyone

4.1 How the loops close

The three components feed each other, which is what makes this a system rather than three scripts:

1. The phantom campaign produces a measured γ, which replaces the literature value in the recommendation engine and in the bandit's action set.
2. Every clinically scored exam updates the acceptability model, and the "Reduce dose" tab then displays the lowest dose proven acceptable for that BMI, drawn from local evidence rather than from physics assumptions.
3. Scored exams whose $\mathrm{CTDI_{vol}}$ and DLP were read from the console are appended to the training data, so the next `python train.py` re-runs the whole model competition on a larger dataset. The dataset grows by itself through routine use.

4.2 Engineering quality

The test suite (13 automated tests) covers the data schema and duplicate grouping, recovery of a known physics law from synthetic data, monotonicity of predictions in mAs and kV, interval consistency, optimiser constraint compliance (including that an obese patient never receives a sub-120 kV proposal), surrogate recovery of a known γ, convergence of the Bayesian optimisation to within 25% of a known optimum, and the bandit's safety and learning behaviour. Training writes a versioned model card and a markdown report for each exam type. The application retrains itself automatically when the installed scikit-learn version differs from the one that produced the stored models.

5 · Results

Question	Answer	Evidence
Can $\mathrm{CTDI_{vol}}$ be predicted from BMI, kV, mAs and length?	**Yes, to 0.9%** (R^2 0.998)	repeated grouped CV on 50 exams
Can DLP be predicted?	**Yes, to 8.2%** (R^2 0.92)	error dominated by the unrecorded scan length
Does deep learning help?	**No** — 2.3% and 12.4%	identical validation protocol for all nine models
Can the tool say what to change?	**Yes** — ranked options	iso-CNR physics plus BMI and tube guard-rails
How much dose could be saved?	**≈40% at 100 kV**	physics estimate; to be confirmed on the phantom
Can the optimum be found experimentally?	**Yes** — ≈8 phantom scans	12 simulated campaigns, within ∼8% of the optimum
Can it keep learning in routine use?	**Yes**, with human approval	1500 simulated patients: 45–50% saving, 3–5% non-diagnostic

WHAT THE SIMULATIONS DO AND DO NOT PROVE

The phantom and bandit results come from simulators in which the true image-quality response is known. They demonstrate that the algorithms converge, respect their constraints and behave safely — that is, they are a test of the software, not clinical evidence. The actual dose reduction at your site depends on your scanner, your reconstruction, your contrast protocol and your readers. Measuring it is precisely what the phantom campaign and the supervised pilot are for.

6 · Questions for the data provider

1. **What exactly is "SCAN SIZE CM"?** It tracks BMI ($r = 0.89$) and does not match the length implied by DLP/$\mathrm{CTDI_{vol}}$ (23.2 cm on average). If the true z-coverage is exportable, the DLP error should fall from about 8% towards the 1% level of the $\mathrm{CTDI_{vol}}$ model.
2. **What differs in rows 2, 9, 22, 41, 45?** They deliver about 7% less $\mathrm{CTDI_{vol}}$ per mAs. Collimation, pitch or scan mode would explain it, and recording that field removes most of the residual $\mathrm{CTDI_{vol}}$ error.
3. **Are rows 28/43 and 29/42 genuinely two patients each?** They are identical in every field.
4. **Is the protocol retrospectively gated?** The dose level strongly suggests it. If prospective triggering is feasible at the patient's heart rate, it is a larger saving than any kV or mAs change — but it is outside the four supplied parameters, so the tool cannot propose it.
5. **Which additional fields can be exported routinely?** In priority order: true scan length; gating mode; collimation and pitch; heart rate; reconstruction kernel and iterative level; and a diagnostic-quality score.

7 · Limitations, safety and roadmap

7.1 Limitations

- **One scanner, one protocol, 50 exams.** The fitted constants k and n belong to this scanner. Another scanner needs its own data — which the "Train / add exams" tab supports, but does not remove.
- **80 and 100 kV are extrapolations.** No exam in the data uses them. The estimates rest on the fitted kV-law and are flagged as such in the interface. The phantom campaign is what turns them into measurements.
- **The image-quality model is first-order.** Noise $\propto 1/\sqrt{\text{dose}}$ and contrast $\propto \mathrm{kV}^{-\gamma}$ ignore beam hardening, iterative reconstruction behaviour, patient shape and cardiac motion — the reasons why the loops measure rather than assume.
- **Dose is not risk.** $\mathrm{CTDI_{vol}}$ and DLP are output metrics, not organ or effective dose. Size-specific dose estimates would need patient diameter.
- **Not a medical device.** No regulatory clearance, no clinical validation. It supports decisions made by qualified staff.

7.2 Suggested next steps

Stage	Action	Outcome
1 · **Data**	Export the additional fields above for the existing exams and for the next 100–200 cases	DLP error towards 1–2%; a model that reflects the real protocol
2 · **Phantom**	One campaign with the physicist: 10–20 scans of a cardiac phantom at the proposed settings, entering measured CNR	Site-specific γ and an evidence-based low-kV protocol
3 · **Pilot**	20–30 patients at the phantom-derived protocol, each scored 1–4 by the reading radiologist, with approval at every step	First real acceptability data; go/no-go decision
4 · **Routine**	Switch on the clinical-learning loop with conservative settings (target 0.97, safety 0.85, 10% steps)	Gradual, evidenced dose reduction; automatic dataset growth
5 · **Expansion**	Add other examinations (brain, chest, abdomen); each gets its own model and report	Department-wide dose-optimisation platform

A · Mathematical summary

A.1 Prediction

$\log \mathrm{CTDI}_{\mathrm{vol}} = a + 1 \cdot \log \mathrm{mAs} + n \log(\mathrm{kVp}/120)$ (Huber loss; a, n free)

$\log \mathrm{DLP} = a' + 1 \cdot \log \mathrm{mAs} + n \log(\mathrm{kVp}/120)$ (constant effective length selected by CV)

Fitted: $k = e^a = 0.0605$ mGy/mAs, $n = 2.53$

A.2 Conformal interval (coverage $1 - \alpha$, exchangeable data)

$r_i = \log y_i - \log \hat{y}_i^{(\text{out-of-fold})}$

$q_{\text{lo}} = \mathrm{Quantile}_{\lfloor (n+1)\alpha/2 \rfloor / n}(r), \qquad q_{\text{hi}} = \mathrm{Quantile}_{\lceil (n+1)(1-\alpha/2) \rceil / n}(r)$

interval $= [\hat{y} e^{q_{\text{lo}}},\ \hat{y} e^{q_{\text{hi}}}]$

A.3 Iso-CNR protocol search

$\mathrm{CNR}_{\mathrm{rel}}(\mathrm{kV}, D) = (\mathrm{kV}_0/\mathrm{kV})^{\gamma} \sqrt{D/D_0} \ \geq \ \tau$

$\Rightarrow \ D_{\text{req}}(\mathrm{kV}) = D_0 \, \tau^2 (\mathrm{kV}/\mathrm{kV}_0)^{2\gamma}, \qquad \mathrm{mAs} = D_{\text{req}} / \big(k(\mathrm{kV}/120)^n\big)$

minimise $\mathrm{DLP} = D \cdot L$ subject to $\mathrm{kV} \geq \mathrm{kV}_{\min}(\mathrm{BMI}),\ \mathrm{mAs} \leq \mathrm{mAs}_{\max},\ L \geq L_{\min}$

A.4 Phantom Bayesian optimisation

surrogate: $\log \mathrm{CNR} = b_0 - \gamma \log(\mathrm{kV}/120) + \delta \log D + \mathrm{GP}(\cdot) + \varepsilon$, priors $\gamma \sim \mathcal{N}(1.35, 0.5^2)$, $\delta \sim \mathcal{N}(0.5, 0.15^2)$

acquisition: $a(x) = \max(0, d^* - d(x)) \cdot \Phi\left(\frac{\mu(x) - \log \mathrm{CNR}_{\mathrm{target}}}{\sigma(x)}\right)$

recommendation: $\arg\min_x d(x)$ subject to $P(\mathrm{CNR}(x) \geq \text{target}) \geq 0.95$

A.5 Safe contextual bandit

$\mathrm{logit}\, P(\text{acceptable}) = w_0 + w_1 \mathrm{BMI}_z + w_2 \log\big(D / D_{\mathrm{ref}}(\mathrm{BMI})\big) + w_3 \log(\mathrm{kV}/120)$

prior $w = (4.0,\ -0.5,\ 6.0,\ -16.0)$, posterior by Laplace approximation

Thompson: $\tilde{w} \sim \mathcal{N}(\hat{w}, \Sigma)$; choose $\arg\min D$ over allowed actions with $P_{\tilde{w}}(\text{acceptable}) \geq 0.97$

allowed $\Leftrightarrow$ $\mathrm{kV} \geq \mathrm{kV}_{\min}(\mathrm{BMI}) \ \wedge \ P_{10\%}(\text{acceptable}) \geq 0.85 \ \wedge \ D \geq 0.85 \cdot$ (lowest proven acceptable at that BMI)

B · Code map and commands

File	Contents
`ctdose/data.py`	PDF/CSV/Excel loading, Greek-and-English header mapping, duplicate grouping, automated audit
`ctdose/models.py`	The nine candidate models, all log-space, physics law with fixed mAs exponent
`ctdose/evaluation.py`	Repeated grouped CV, metrics, 1-SE selection, conformal calibration
`ctdose/predictor.py`	Trained bundle: prediction, intervals, applicability domain, peer quantiles, mAs inversion
`ctdose/optimizer.py`	Iso-CNR search, guard-rails, ranked options, narrative recommendations
`ctdose/phantom_bo.py`	CNR surrogate, constrained expected improvement, campaign persistence, simulator
`ctdose/bandit.py`	Bayesian logistic model, Thompson sampling, safety rules, logging, reader simulator
`app.py · train.py · tests/`	Streamlit interface (7 tabs), training and reporting pipeline, 13 automated tests

```
pip install -r requirements.txt  ·  python app.py  ·  python train.py  ·  python -m pytest
-q
```

C · References

1. Stocker TJ et al. *Reduction in radiation dose in cardiovascular computed tomography imaging: results from the PROTECTION VI international, multicentre, observational study.* European Heart Journal, 2018 — source of the 400 mGy·cm reference level for CCTA and of the kV-by-BMI recommendation (80 kV below BMI 25, 90–100 kV for 25–30).
2. *Radiation Doses in Cardiovascular Computed Tomography* (review, PMC10141413) — compilation of national CCTA diagnostic reference levels: United Kingdom 170 (prospective) / 380 (retrospective), France 393 / 870, Australia 268, Saudi Arabia 395 / 1057 mGy·cm.
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5. Gelbart MA, Snoek J, Adams RP. *Bayesian Optimization with Unknown Constraints*, UAI 2014.
6. Sui Y, Gotovos A, Burdick J, Krause A. *Safe Exploration for Optimization with Gaussian Processes* (SafeOpt), ICML 2015 — the safe-expansion principle used by the bandit's step limit.